Implementing Self-Registration with Flows

Introduction

Flows, introduced in WSO2 Identity Server 7.2.0 and available in WSO2 Identity Server 7.3.0, is a powerful framework for designing fully customizable user interaction journeys.

Flows allow administrators and developers to define step-by-step processes for identity-related actions without writing custom code. Using flows, you can visually compose secure, guided experiences for:

1. Self Registration
2. Password Recovery
3. Invited User Registration

Scenario

You are an administrator of Guardio Insurance. New customers must be able to create their own accounts. After a customer fills in the sign-up form, they choose how to prove that the email address belongs to them:

- **Email OTP**: Identity Server emails a one-time code that the customer types in.
- **Magic Link**: Identity Server emails a link that the customer clicks.

The account is created only after the email address is verified. You build this journey in the Flows designer, without writing code.

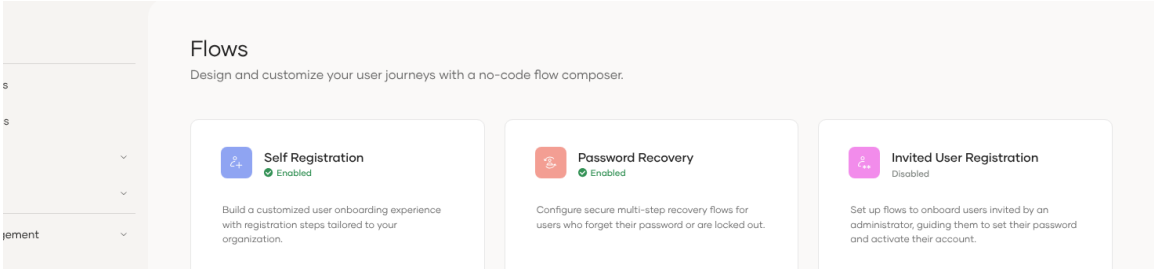
Prerequisites

- WSO2 Identity Server 7.3.0 is running and you can log in to the Console as admin.
- The email server is configured (see Lab 03, Configure the email server), and you can open the inbox of a test email address.
- If you enabled self-registration in Lab 02, this lab replaces that registration flow.

Part 1: Build the registration flow

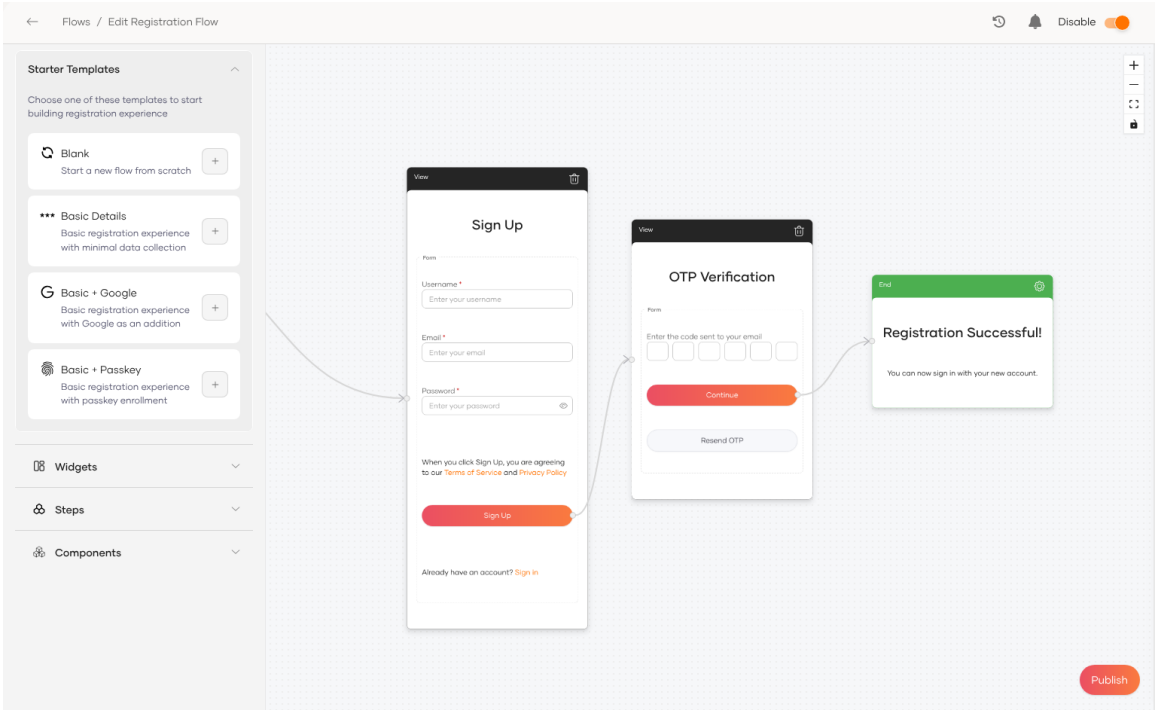
Step 1: Open the flow designer

1. In the WSO2 Identity Server Console, go to **Flows**.
2. Click the **Self Registration** card.



Step 2: Start from the Basic Details template

1. In **Starter Templates**, click the **+** next to **Basic Details**. (On a new flow it is already loaded.)
2. The canvas shows three steps: the **Sign Up** view (username, email and password), an **OTP Verification** view, and the **End** step.



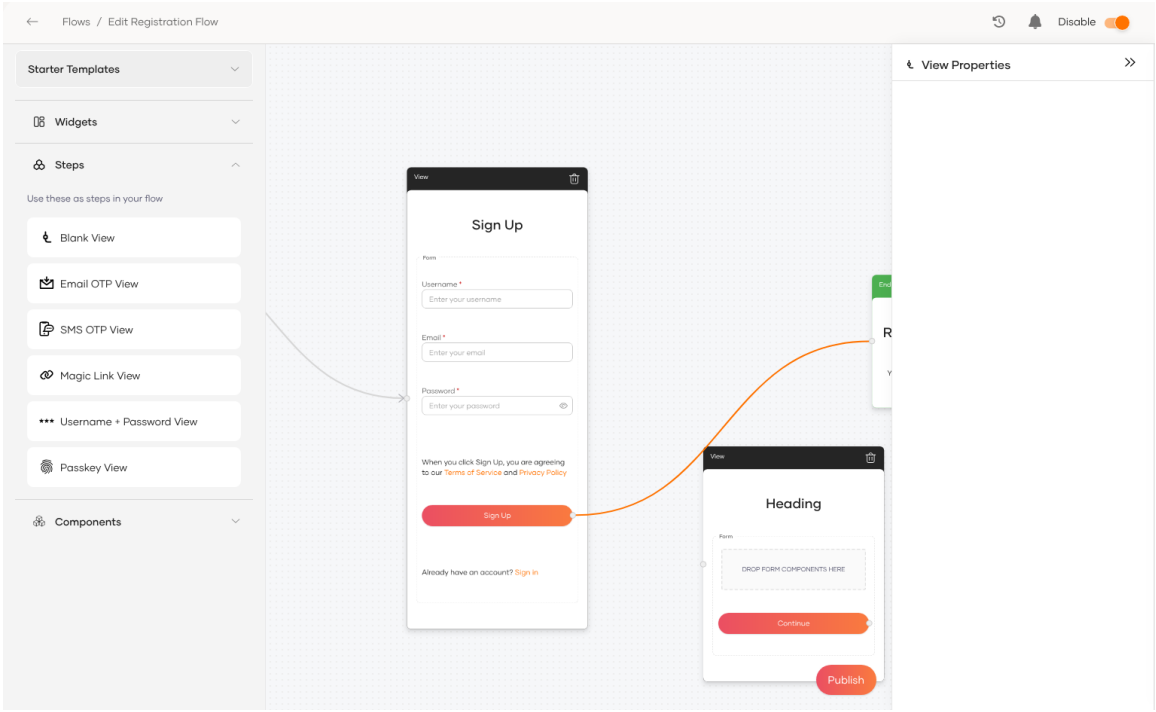
Step 3: Remove the OTP Verification view

1. Click the bin icon on the header of the **OTP Verification** view. You will add the verification steps again in Step 5, as a choice.

Step 4: Add a view where the user chooses how to verify

1. Collapse **Starter Templates** and expand **Steps**.

2. Drag **Blank View** onto the canvas, to the right of the Sign Up view. It contains a **Heading**, an empty form (**DROP FORM COMPONENTS HERE**) and a **Continue** button.

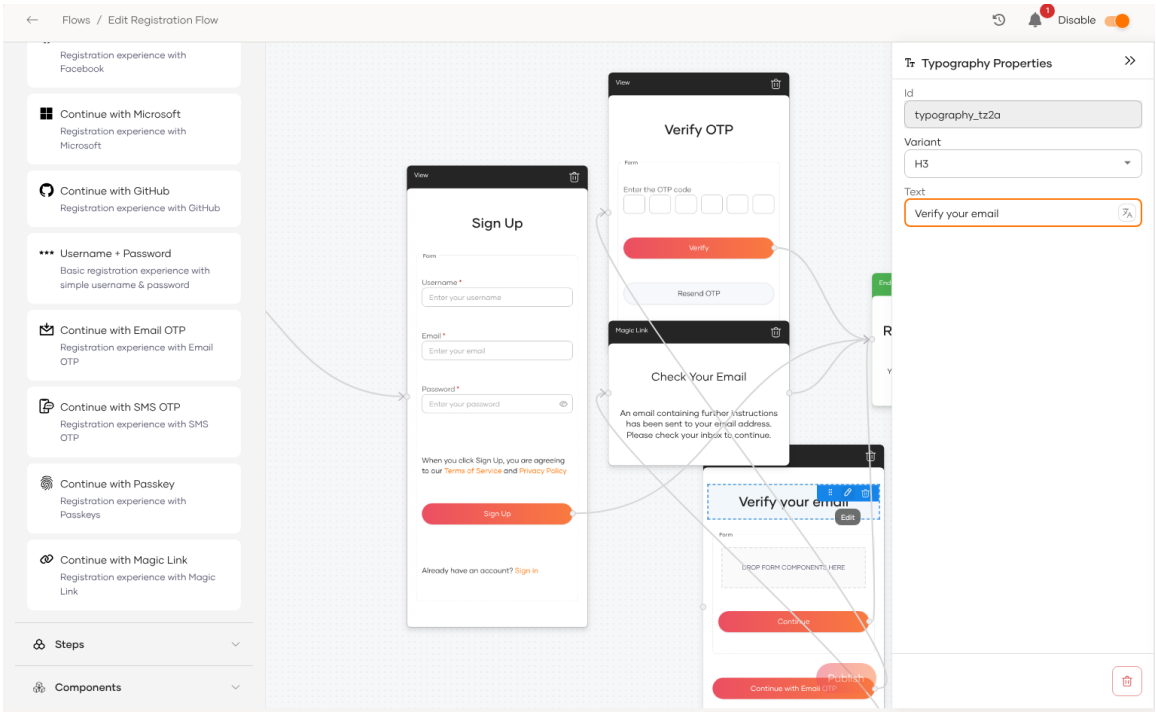


Step 5: Add the two verification options

1. Collapse **Steps** and expand **Widgets**.
2. Drag **Continue with Email OTP** onto the **DROP FORM COMPONENTS HERE** area of the new view. This adds a **Continue with Email OTP** button and creates a **Verify OTP** view that is already connected to **End**.
3. Drag **Continue with Magic Link** onto the same area. This adds a **Continue with Magic Link** button and creates a **Magic Link** (Check Your Email) step, also connected to **End**.

Step 6: Tidy the choice view

1. Hover over **Heading** and click the pencil icon (**Edit**). In the properties panel, change **Text** to `Verify your email`.



2. Hover over the **Continue** button in the same view and click the bin icon (**Delete**). Otherwise users could continue without verifying their email.

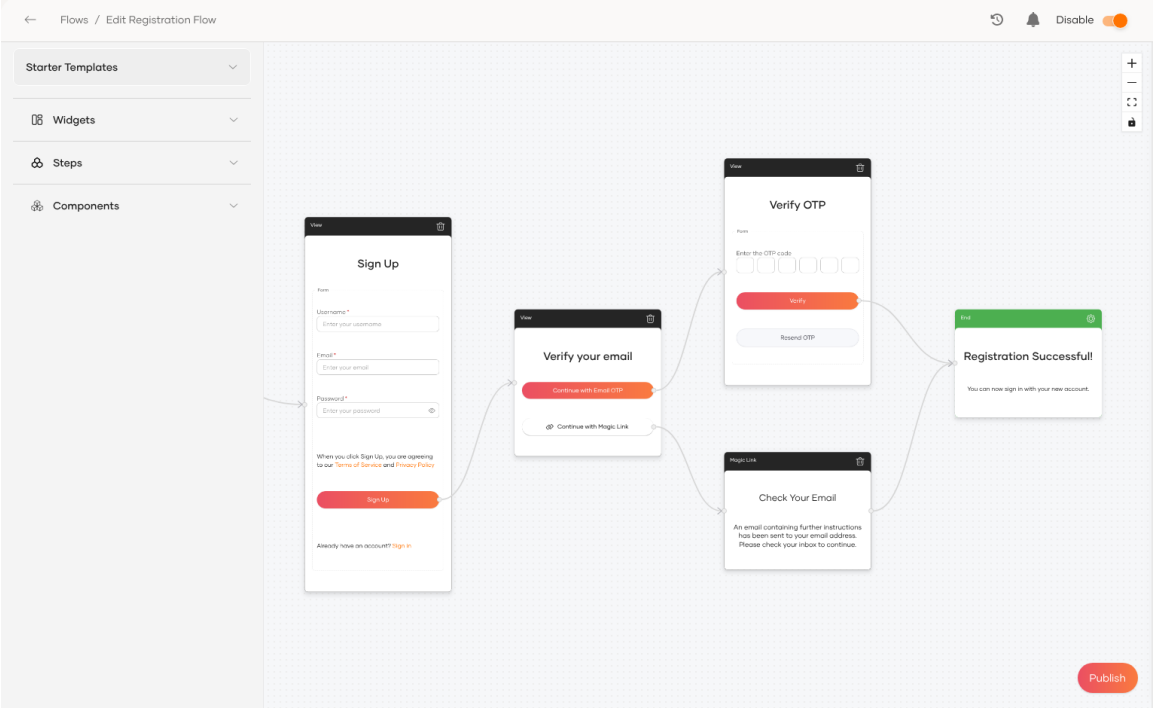
3. Hover over the empty **Form** and click the bin icon to delete it as well.

Step 7: Connect the Sign Up button to the choice view

1. Drag from the small dot on the right edge of the **Sign Up** button to the left edge of the **Verify your email** view. A line now connects them.

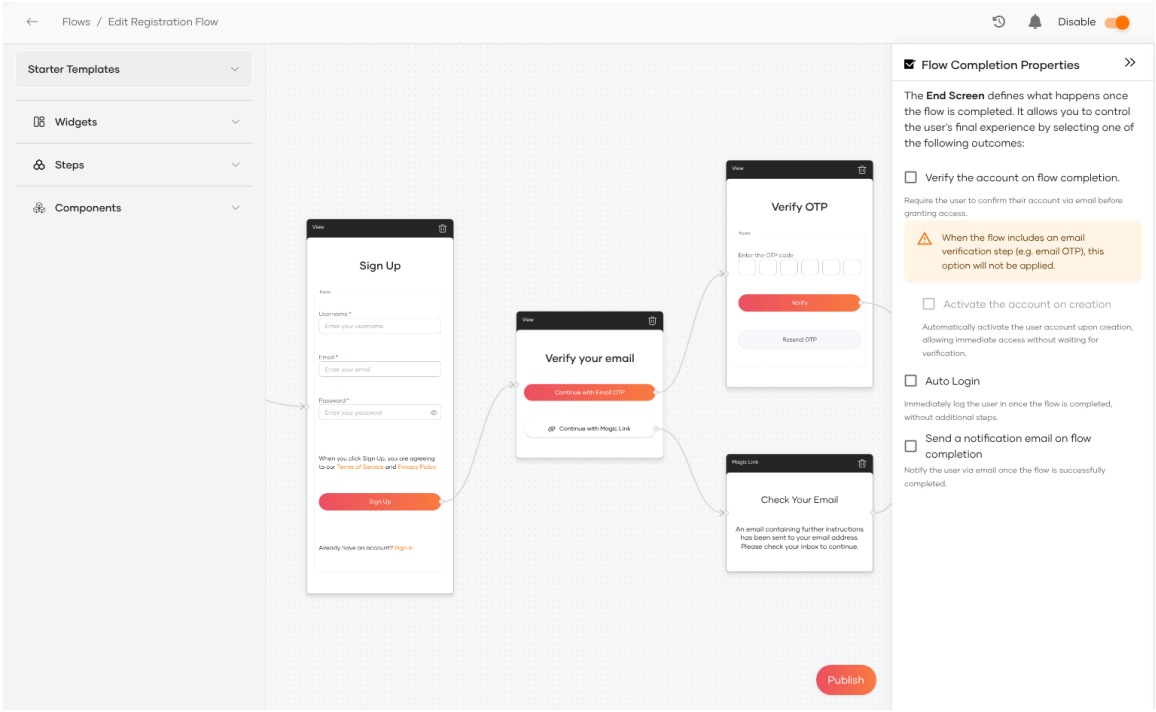
2. Click the bell icon at the top. It should show **No errors found**. (If it shows "Email OTP requires at least one email field to be present in the flow", the Sign Up button is not connected to the Verify your email view yet.)

   The finished flow looks like this (drag the views to arrange them):



Step 8: Set what happens when the flow completes

1. Click the gear icon on the **End** step to open **Flow Completion Properties**.
2. Untick **Verify the account on flow completion**. Both options already verify the email address. If you leave it ticked, users who choose Magic Link get an extra "Confirm your account" email and cannot sign in until they click it.



Step 9: Publish the flow

1. Click **Publish** at the bottom right. (If the flow is not enabled yet, the button is called **Save draft**: click it, then turn on **Enable** at the top right.)

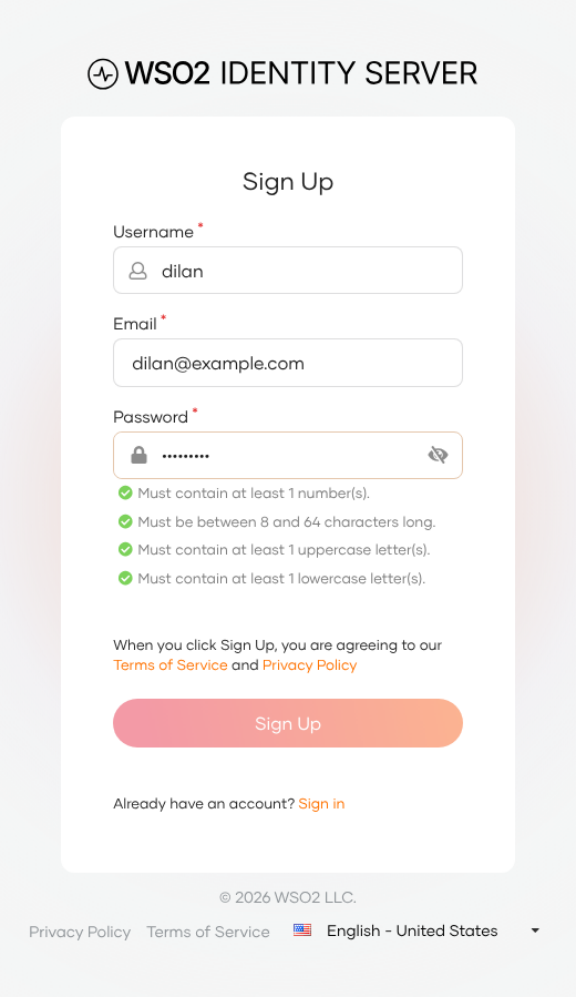
2. The switch at the top right shows **Disable**, which means the flow is enabled.

Part 2: Try it

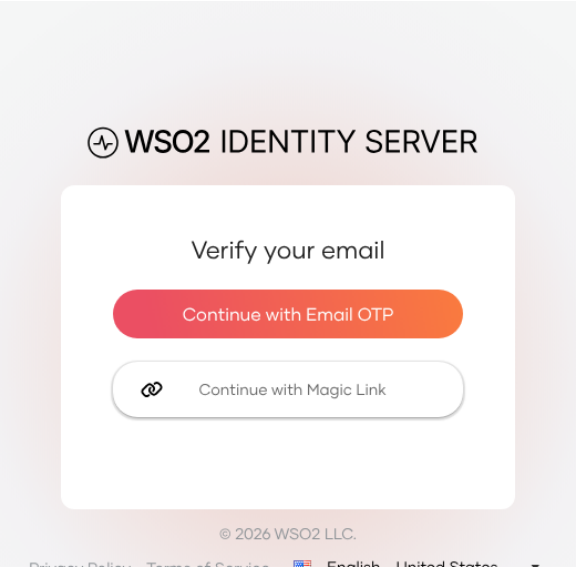
A. Register and verify with Email OTP

1. Open a private browser window and go to `https://localhost:9443/myaccount`. On the sign-in page, click **Register**.

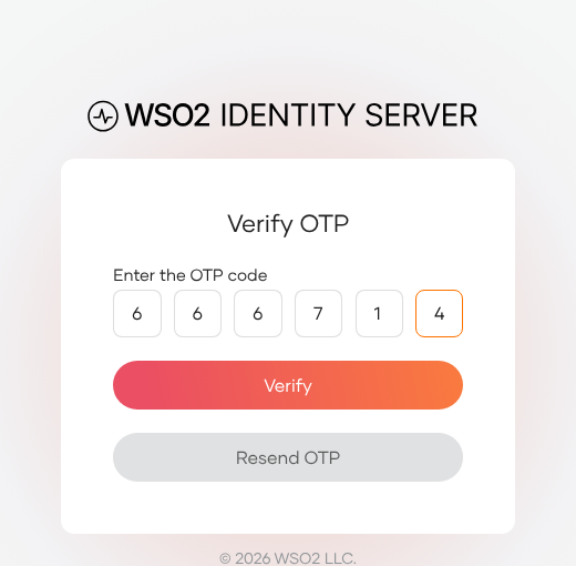
2. Enter a username, an email address you can open, and a password that meets the rules shown, then click **Sign Up**.



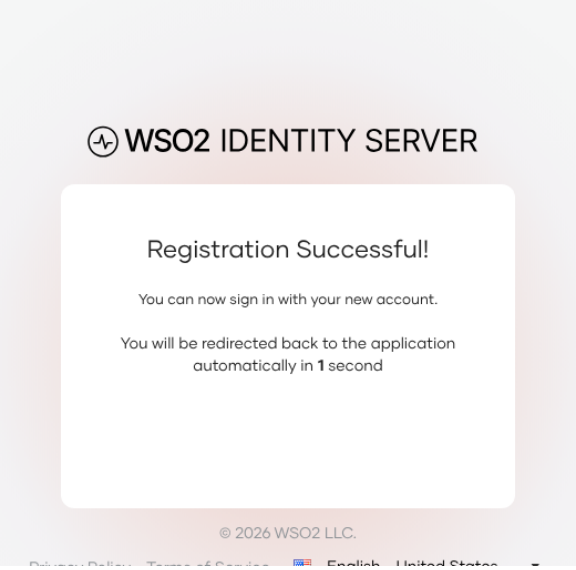
3. On the **Verify your email** page, click **Continue with Email OTP**.



4. Open the email **Verify your email to complete registration**, enter the code on the **Verify OTP** page and click **Verify**.



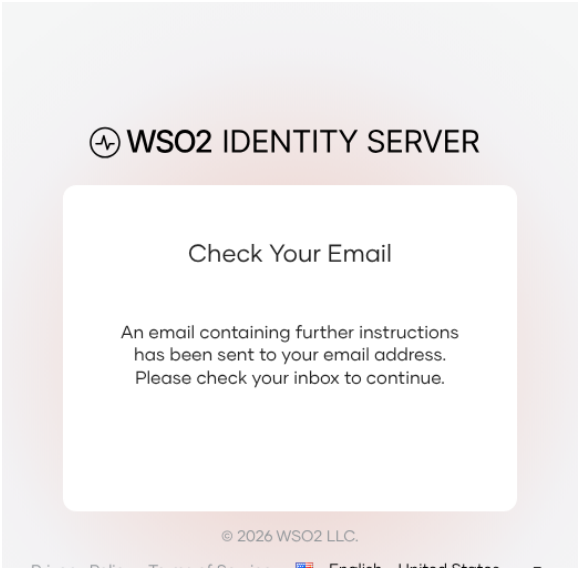
5. The **Registration Successful!** page appears and you are sent back to My Account. Sign in with the new username and password.



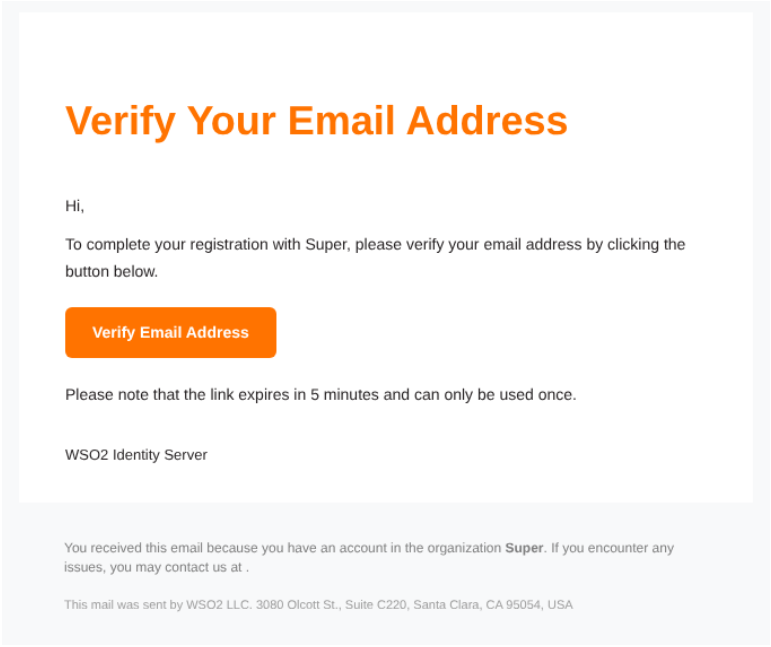
B. Register and verify with Magic Link

1. In a new private window, start the registration again with a different username and email address.

2. On the **Verify your email** page, click **Continue with Magic Link**. A **Check Your Email** page appears.



3. Open the email **Verify your email to complete registration** and click **Verify Email Address**. Open it in the same browser; the link works only once and expires after 5 minutes.



4. You are taken to the sign-in page. Sign in with the new username and password.

C. Check the new accounts

1. In the Console, go to **User Management > Users**. Both new users are listed.

Expected outcome

- New users can register themselves from the sign-in page.
- Each user chooses Email OTP or Magic Link to verify the email address.
- The account is created only after verification, and the user can sign in straight away.